

The Ratepayer-Centered Data Center Incentive Framework (RCD-CIF): Aligning Advanced Computing Architectures and Baseload Renewables with Public Value

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Table of Contents

Abstract	2
Description	2
1. Research on the Need.....	3
2. Technical Component 1: Hardware Efficiency (Process-in-Memory).....	3
3. Technical Component 2: Geothermal Baseload Renewable Power	4
4. Technical Component 3: Socio-Economic Value.....	5
5. The RCD-CIF Scoring Mechanism.....	5
6. Discussion on Weights	6
7. The Incentive Structure	7
8. Schematic Overview of the Design.....	8
9. Implementation & Policy Recommendations	10
10. Limitations of the RCD-CIF Framework.....	10
11. Problem Tree	11
12. Next Steps	13
References	15

Abstract

This project develops the Ratepayer-Centered Data Center Incentive Framework (RCD-CIF), a performance-based model that aligns public tax incentives with measurable technological, environmental, and socio-economic outcomes. As data centers expand and place increasing stress on energy systems and communities (International Energy Agency, 2023; National Renewable Energy Laboratory, 2024), many states continue offering subsidies without evaluating efficiency or public benefit (Good Jobs First, 2025; Green & Nguyen, 2025). The RCD-CIF addresses this gap through three evaluation domains: hardware efficiency metrics emphasizing Processing-in-Memory architectures, renewable energy performance centered on geothermal baseload power, and socio-economic indicators such as job creation and fiscal return. These components feed into a unified scoring system that assigns facilities to incentive tiers. Through iterative design and analysis, the project transforms a conceptual idea into a practical framework supporting sustainable, ratepayer-aligned data center development.

Description

Our final project brings to life the concept developed during the midterm proposal, a complete prototype of the Ratepayer-Centered Data Incentive Framework (RCD-CIF). The RCD-CIF is designed to address the gap between the unprecedented energy demands of modern data centers and the public subsidies that support them. Today, most state-level tax incentive structures treat all data centers equally, regardless of whether they impose extreme strain on local grids or meaningfully contribute to community well-being. This “blind subsidy” approach has been repeatedly criticized by public utility commissions, economic development agencies, and energy researchers. Large data center load growth, especially without accompanying planning, can strain grid infrastructure, a concern documented in grid-planning analyses by NREL and others (NREL, 2024; NREL Chip-to-Grid Initiative, 2025)

To bring our vision into reality, we created a performance-driven framework that ties public incentives to measurable engineering, environmental, and socio-economic outcomes. The final work product is a comprehensive system with four major components: 1) A technical metric suite evaluating next-generation computing hardware such as Process-in-Memory (PIM); 2) A renewable energy integration assessment, with emphasis on geothermal baseload power; 3) A socio-economic return model that evaluates job creation, fiscal return, and community benefits; 4) A unified scoring and incentive mechanism that allocates tax benefits proportionally to public value created.

Together, these components form a policy tool that state agencies could adopt immediately to differentiate between high-value and low-value data center projects.

1. Research on the Need

Data centers now represent one of the fastest-growing loads on U.S. electrical grids. The International Energy Agency (IEA) estimates that AI and cloud workloads could triple global data center energy consumption by 2030 (IEA, 2023). Traditional GPU-based computing systems are increasingly inefficient due to the Von Neumann bottleneck, the large energy cost associated with moving data between memory and compute units (Hennessy & Patterson, 1989). As AI model sizes increase exponentially, this bottleneck becomes the dominant driver of power consumption.

At the same time, many states continue to subsidize new data centers without requiring them to source firm clean energy or provide demonstrable community benefits (Green & Nguyen, 2025; Good Jobs First, 2025). This mismatch between public cost and public value is the central problem the RCD-CIF was designed to solve.

Our final work product demonstrates how a performance-based model can correct this misalignment while accelerating adoption of advanced, sustainable technologies.

2. Technical Component 1: Hardware Efficiency (Process-in-Memory)

Process-in-Memory (PIM) represents one of the most promising approaches to increasing compute efficiency for AI workloads. Unlike traditional architectures, PIM integrates processing units directly within or adjacent to memory arrays, dramatically reducing the energy required for data movement. Samsung Electronics reports that commercial HBM-PIM systems reduce energy per bit transferred by over 70% compared to GDDR6 (Samsung, 2021). Academic studies demonstrate similar results, showing order-of-magnitude improvements in energy efficiency and bandwidth utilization (Ahn et al., 2015).

To operationalize these insights, our team developed the following hardware metric table, representing the technical parameters we included in the RCD-CIF scorecard.

Table 1. Hardware Efficiency Metrics (PIM)

Goal: Measure the reduction in data movement energy, which is the primary driver of AI energy bloat.

Parameter	Metric	Benchmark (Tier 1 Legacy)	Target (Tier 4 PIM)	Why It Matters
Data Movement Energy	pJ/bit (Picojoules per bit)	> 10 pJ/bit (GDDR6)	< 1.5 pJ/bit (HBM-PIM)	Reduces total MW load (Samsung, 2021).

System Energy Efficiency	TOPS/W (Tera-Operations per Watt)	~2-4 TOPS/W	~10-15 TOPS/W	Higher compute per watt improves capacity (Ahn et al., 2015).
Memory Bandwidth Utilization	% Effective Bandwidth	~30-40% (Von Neumann Bottleneck)	> 80%	Eliminates idle cycles and wasted energy.

3. Technical Component 2: Geothermal Baseload Renewable Power

Unlike intermittent renewables such as solar and wind, geothermal energy provides firm, 24/7 power with capacity factors above 85–95% (U.S. Department of Energy [DOE], 2022). This stability eliminates the need for fossil-fuel peaker plants, which utilities typically deploy when intermittent resources fluctuate.

Geothermal direct-use systems also significantly reduce water consumption compared to traditional evaporative cooling. An important consideration for regions facing water scarcity. According to DOE lifecycle analyses, closed-loop geothermal cooling uses over 90% less water per kWh than typical evaporative systems (DOE, 2022).

To capture these advantages, our solution includes geothermal-focused metrics.

Table 2. Geothermal Power Metrics

Goal: Measure the stability and cleanliness of the power supply.

Parameter	Metric	Benchmark (Tier 1 Grid Mix)	Target (Tier 4 Geo)	Why It Matters
Capacity Factor	% Availability (24/7)	Variable (Solar ~25%, Wind ~35%)	> 90% (Baseload)	Ensures reliability and reduces strain (DOE, 2022).
Carbon Intensity	gCO ₂ eq/kWh	~400g (Grid avg)	< 20g	Genuine decarbonization (NREL, 2022).
Water Usage Effectiveness	WUE (L/kWh)	> 1.8 L/kWh (Evaporative)	< 0.1 L/kWh (Closed-Loop)	Protects aquifers (DOE, 2022).

4. Technical Component 3: Socio-Economic Value

Many data centers generate relatively few permanent jobs compared to their size, while construction phases employ many workers; ongoing staffing is often minimal once the facility is operational (Tozzi, 2025). However, geothermal plants require ongoing operations and maintenance of staff, increasing local employment.

Also, studies show that many data center incentive packages result in negative fiscal ROI, returning less in tax revenue than was provided in subsidies (Bartik, 2019). Our framework incentivizes facilities to generate *positive* fiscal outcomes.

Table 3. Socio-Economic Impact Metrics

Goal: Measure the "Return on Subsidy" for the community.

Parameter	Metric	Benchmark (Tier 1)	Target (Tier 4)	Why It Matters
Job Multiplier	FTE Jobs per MW	< 0.5 (Highly Automated)	> 2.0 (Geo Plant Staff + IT)	Geothermal plants require permanent O&M staff; simple server farms do not.
Fiscal Velocity	Local Tax \$ Generated / \$ Incentivized	< \$0.80 (Negative ROI)	> \$1.50 (Positive ROI)	Ensures taxpayers profit (Bartik, 2019).
Community Benefit	District Heat Sharing (Yes/No)	No (Heat Vented to Air)	Yes (Heat Recycled)	Using waste heat for local homes/greenhouses adds tangible public value.

5. The RCD-CIF Scoring Mechanism

The RCD-CIF converts the metrics above into a unified quantitative score:

$$RCD\ Score = (0.40 \times PIM\ Factor) + (0.30 \times Geo\ Factor) + (0.30 \times Socio\ Factor)$$

This weighting was selected after reviewing industry literature on the relative impact of compute efficiency, energy source stability, and socio-economic contribution (IEA, 2023; DOE, 2022).

6. Discussion on Weights

The RCD-CIF utilizes a 40:30:30 weighted scoring model. We derive it through a Cost-Benefit consideration of Grid Impact:

40% Hardware Efficiency: We prioritize hardware efficiency highest because demand-side reduction offers the highest return on investment for the grid. Reducing the thermal design power (TDP) of the facility lowers both the electrical load and the water-cooling requirement simultaneously. Economically, it is more efficient to incentivize the reduction of 1 MW of demand than to subsidize the generation of 1 MW of new supply.

30% Renewable Integration: This weight reflects the market premium of baseload renewable power. Unlike intermittent wind or solar, which often require the utility to maintain fossil-fuel backup generation, Geothermal Direct Use provides firm capacity. This component rewards data centers that do not externalize reliability costs onto the public grid.

30% Socio-Economic: This weighting serves as the fiscal control mechanism. It ensures that the value of the tax abatement provided by the state is offset by tangible economic multipliers, specifically local job creation and secondary community infrastructure (district heating). This prevents ‘hollow’ investments where public subsidies exceed the long-term fiscal receipt.

The 'Loading Order' framework established by the California Energy Commission and the EPA, which mandates that demand-side efficiency (represented here by the 40% Hardware weight) must be prioritized over new generation capacity (the 30% Geothermal weight) has also been referenced. Furthermore, this distribution reflects a Swing Weighting approach consistent with Multi-Criteria Decision Analysis (MCDA), where the highest weight is assigned to the variable with the greatest variance in performance outcomes—in this case, the 70% energy reduction potential of PIM technology.

In addition, we attribute the relative weighting in the RCD-CIF using standard welfare- and cost-effectiveness. First, the 40% weight on hardware efficiency (e.g., Processing-in-Memory architectures) relative to the 30% weight on geothermal supply has been attributed in the distinction between the levelized cost of electricity (LCOE) and the levelized avoided cost of electricity (LACE), a valuation framework used by the U.S. Energy Information Administration to compare the cost of producing electricity with the value of avoided generation and capacity needs. Under this framing, geothermal LCOE represents the unit cost of delivering additional low-carbon electricity, while efficiency improvements function as demand-side resources whose value is realized through avoided generation, avoided transmission and distribution expansion, and reduced fuel consumption. Because demand-side efficiency (i.e. ‘negawatts’) often delivers energy services at a lower cost per MWh than the cost of supplying new firm generation, prioritizing hardware efficiency with a modest premium is economically defensible: a one-unit improvement in computing efficiency can reduce total system costs through compounding effects (lower IT load,

lower cooling load, and reduced peak contribution), whereas new clean generation primarily addresses the supply side of the same load.

Second, the equal 30% weights on geothermal integration and socio-economic outcomes can be framed as an intentional balancing of global environmental externalities and local economic returns. In benefit–cost terms, geothermal adoption generates a global public benefit through avoided emissions, which can be monetized using the Social Cost of Carbon (SCC) as a shadow price for CO2 reductions. In parallel, socio-economic performance such as employment, tax base expansion, local spillovers) can be evaluated using local fiscal return concepts and regional multiplier effects commonly used in economic development analysis. A stylized calibration illustrates why parity between these categories is reasonable: for a large facility, switching from fossil-based electricity to geothermal can avoid a substantial volume of annual emissions whose SCC-valued benefit can reach the same order of magnitude as the local economic activity and fiscal receipts generated by a modest but high-quality employment footprint plus tax contributions. Accordingly, by assigning equal weights to clean firm supply and community return, we aim to reflect on a normative policy objective: incentives should internalize the global climate externality while also meeting a ‘ratepayer legitimacy’ constraint that public subsidies yield tangible local value rather than functioning as unconditional abatements.

7. The Incentive Structure

The final score maps to a tiered incentive structure.

Table 4. Incentive Tier Structure

The policy output: Dynamic tax rates based on the Audit Score.

RCD Score	Quadrant Classification	State Sales Tax Exemption	Property Tax Abatement	Permitting Speed
85 - 100	Tier 4: Symbiotic Partner	100% Exemption	100% for 20 Years	Expedited (Fast-Track)
70 - 84	Tier 2: Good Neighbor	60% Exemption	50% for 10 Years	Standard
50 - 69	Tier 3: Grid Battery	30% Exemption	ITC Only (CapEx Credit)	Standard
< 50	Tier 1: Parasitic Load	15% Exemption	0% Abatement	Subject to Impact Fees

8. Schematic Overview of the Design

To elucidate the final implementation, we provide three figures that visually summarize the solution’s design, operational flow, and resulting state, offering a clear and consolidated graphical reference.

Figure 1 illustrates the overall structure of the Ratepayer-Centered Data Center Incentive Framework (RCD-CIF). The diagram shows how the evaluation system integrates three major metric categories: hardware efficiency, energy source quality, and socio-economic impact. Each category captures a distinct dimension of data center performance and feeds into a unified RCD score ranging from 0 to 100 points. The score is then used to determine the appropriate incentive tier, linking public benefits directly to demonstrated performance.

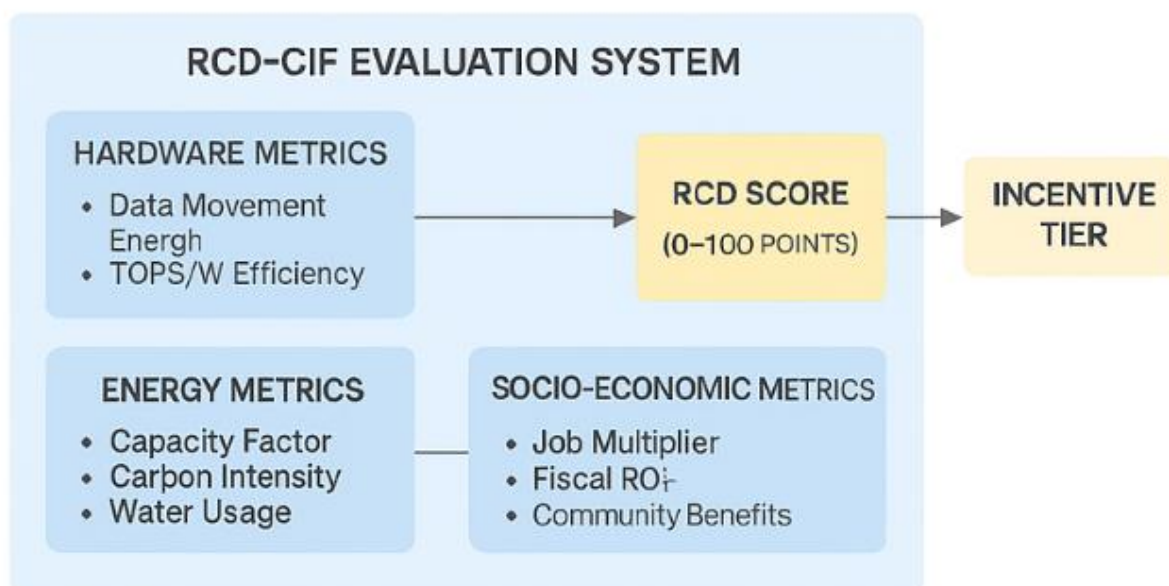


Figure 1. RCD-CIF Architecture Overview

Figure 2 presents the RCD-CIF scorecard, a standardized tool used to evaluate data center proposals. The scorecard organizes metrics according to the framework’s three weighted categories: Hardware Efficiency (40%), Energy Source (30%), and Socio-Economic Value (30%). The total score out of 100 determines a facility’s incentive eligibility. This scorecard serves as the

operational format that policymakers or regulatory agencies would use during annual audits or initial project assessments.

Figure 3 visualizes the classification of data centers across two critical dimensions: grid impact and socio-economic multiplier. The quadrant framework categorizes facilities into four profiles, Symbiotic Partner, Good Neighbor, Grid Battery, and Parasitic Load, based on how beneficial or detrimental they are to both the electrical grid and the surrounding community. This diagram communicates the conceptual logic behind the tier system and clarifies how qualitative performance characteristics map onto incentive classifications.

RCD-CIF SCORECARD	
Facility Name:	_____
Year:	_____
HARDWARE EFFICIENCY (40%)	
PIM Adsption %	Score: <u>80</u>
Bandwidth Utilization	_____
ENERGY SOURCE (30%)	
% Geothermal Baseload	Score: <u>30</u>
Carbon Intenaity	Score: <u>30</u>
TOTAL SCORE:	<u>/100</u>
TIER CLASSIFICATION	

Figure 2. RCD-CIF Scorecard (left)

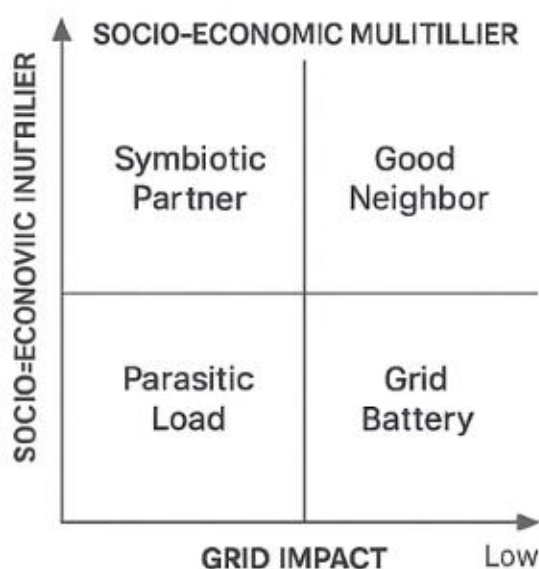


Figure 3. Socio-Economic $\times$ Grid Impact Quadrant (right)

9. Implementation & Policy Recommendations

To ensure accountability, and to ensure the RCD-CIF is realistic for policymakers and implementable at scale, the final work product includes the following three components.

1. The Clawback Clause, widely recommended in economic development policy (Bartik, 2019). If a facility (Tier 4) turns off its PIM hardware or switches to grid power (dropping its score below 85), the difference in tax savings can be "clawed back" by the state.
2. The Transition Grant enables retrofits of legacy GPUs to PIM architectures. A specialized grant fund to help Tier 1 legacy data centers retrofit with PIM hardware to move up the quadrant.
3. Annual audits aligned with international sustainability reporting standards (Global Reporting Initiative, 2021).

10. Limitations of the RCD-CIF Framework

While the RCD-CIF offers a robust mechanism for aligning private investment with public value, several structural and implementation limitations must be acknowledged.

The framework's heavy reliance on geothermal energy (30% weight) creates a geographic bias. High-quality geothermal resources are not uniformly distributed across the United States; they are predominantly concentrated in the Western states such as Nevada, California). Consequently, the 'Tier 4: Symbiotic Partner' designation may be geologically unattainable for data centers in resource-poor regions including the Southeast or Midwest, regardless of their willingness to invest. This could inadvertently penalize efficient facilities simply due to their location, necessitating a potential 'Geographic Adjustment Factor' in future iterations of the scoring model.

The hardware efficiency metrics assume the commercial availability and seamless integration of Processing-in-Memory technologies. Currently, PIM is primarily in the advanced prototype or early commercial adoption phase for example Samsung HBM-PIM. There is a risk of a potential hardware gap where policy incentives are enacted before the supply chain can support widespread PIM deployment. Furthermore, retrofitting legacy data centers with PIM hardware—while incentivized by the proposed 'Transition Grant' presents significant software compatibility challenges that may slow adoption rates more than financial incentives alone can overcome.

The proposed annual audit mechanism requires state agencies to verify highly technical metrics, such as Picojoules per bit and Memory Bandwidth Utilization. Most state revenue departments or economic development agencies lack the in-house engineering expertise to validate these claims accurately. Without third-party verification bodies or standardized industry reporting protocols,

the framework faces a high risk of information asymmetry, where operators could manipulate technical reporting to secure higher incentive tiers.

The relative weighting ratio is a static heuristic. It does not account for the diminishing marginal returns of efficiency. For example, once the grid fully decarbonizes the value of ‘avoided carbon drops, while the value of grid stability might rise. The current RCD-CIF thus may lack a logical automated ratcheting mechanism that dynamically adjusts these weights in response to changing grid carbon intensity or local labor market conditions.

By imposing stricter efficiency and renewable mandates, the RCD-CIF could theoretically drive data center development to neighboring states with ‘blind subsidy’ models (Tier 1 policies). While this protects the implementing state's grid, it may result in economic leakage where jobs and tax revenue migrate across state lines while the regional carbon emissions remain unchanged. This suggests that the RCD-CIF would be most effective if adopted as a multi-state regional compact rather than a single-state policy.

11. Problem Tree

The RCD-CIF framework was developed through a mix of Divergent and Convergent Thinking, guided by curiosity, grounded in research, and shaped by interdisciplinary exploration.

We proceeded to Problem Tree Visualization. The main issue, “High energy consumption and heat buildup in data centers”, was broken down into underlying contributors such as High Data Movement, Poor Heat Dissipation, and Excess Electricity Usage. This helped pinpoint where innovations could have the most impact.

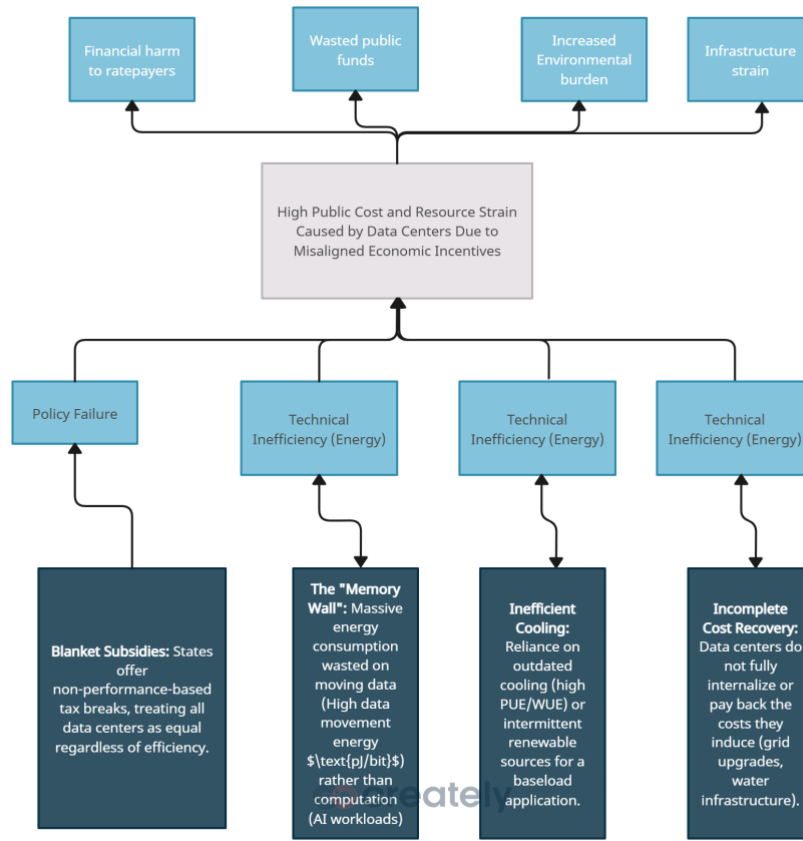


Figure 4. Problem Tree

In addition, a 2x2 Matrix was created to compare four solution pathways, combining Traditional vs. PIM computation and Traditional vs. Geothermal cooling. Tier 4: PIM + Geothermal emerged as the most efficient and high-impact pathway. To record the curiosity process, each team member kept logs with questions on PIM limitations, geothermal seasonal performance, and the conditions required to justify incentives. These logs helped maintain research focus and continuity. After comparing all alternatives, the team converged on a unified, performance-based policy integrating PIM and geothermal cooling due to their combined energy savings, feasibility, and mutual reinforcement.

12. Next Steps

Step	Key Actions	Timeline	Resources Needed	Who Needs to Act/Talk To
1. Direct Policy Action	Submit the RCD-CIF as an official public comment to key utility regulatory bodies (e.g., Illinois Commerce Commission).	1–3 Months	Legal/Policy drafting support for formal submission.	State Public Service Commissioners (PSCs), Utility Regulatory Staff, General Counsels.
2. Coalition Building	Work with ratepayer advocacy organizations (e.g., AARP, local community groups) and environmental/policy groups to build public support.	3–6 Months	Communications team, Grassroots organizing, Economic modeling for local impact.	Non-profit Ratepayer Advocates, Environmental Policy Think Tanks, Local Elected Officials.
3. Model Legislation	Develop ready-to-introduce bill language (Model Legislation) for state legislatures. This would include provisions for a ratcheting PUE target.	6–12 Months	Legislative drafters, Policy analysts to forecast PUE improvement rates.	State Legislators, Legislative Counsel, Economists.
4. Industry Dialogue	Advance the PIM and Geothermal	Ongoing	Technical subject matter experts	Data Center Operators (e.g.,

	technical standards through dialogue with manufacturers and data center operators.		(PIM/Geothermal), Industry association contacts (e.g., iMasons, The Green Grid).	National Petascale Computing Facility, Amazon, Microsoft, Meta), Hardware Manufacturers (e.g., Samsung, UPMEM).
5. Pilot Program Launch	Identify a state/region with an active data center build-out to implement a limited-scope RCD-CIF pilot program.	12–24 Months	Securing state legislative funding, Project management, Third-party auditor for performance verification.	State Economic Development Agencies, Utility Companies, Pilot Data Center Operator.

Table 5. Next Steps

To keep the policy relevant over time, the framework should include a built-in system for regularly updating its standards. Every few years (for example, every 2-3 years), the performance tiers should automatically tighten based on industry-wide data. This ensures the policy continues to push innovation forward and prevents top subsidy levels from being awarded for performance that will soon be considered average.

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